Problems with Group Decision Making

Comparative Politics

November 8-12, 2021

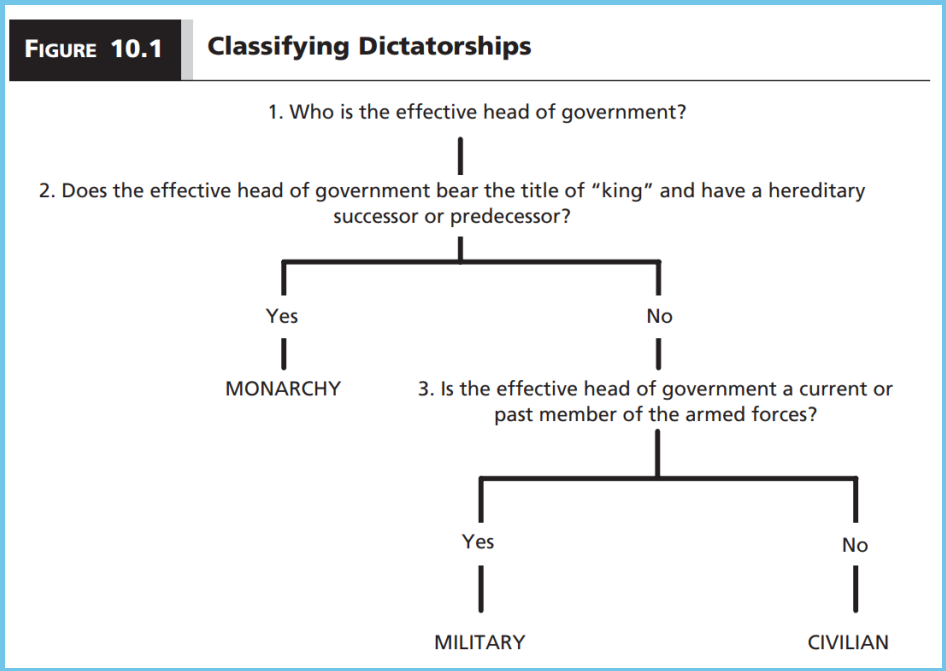
Last time

- We explored whether democracy or dictatorship would be the most desirable regime based on their ability to guarantee material well-being
- We concluded that democracy is **sufficient** for material well-being
- But also pointed out that many improvements to promote well-being in democracies happened **before democratization**
- This is a **consequentialist** approach to the democracy vs. dictatorship debate
- **This week:** Take a **deontological** approach

Deontological approach

- Before, we evaluated regimes based on whether they lead to good outcomes `(material well-being)`
- Now, the question is whether any regime intrinsically desirable properties
- How do different regimes make decisions? Is any regime better at decision-making?
- We will focus on how democracies make decisions

We are skipping dictatorships

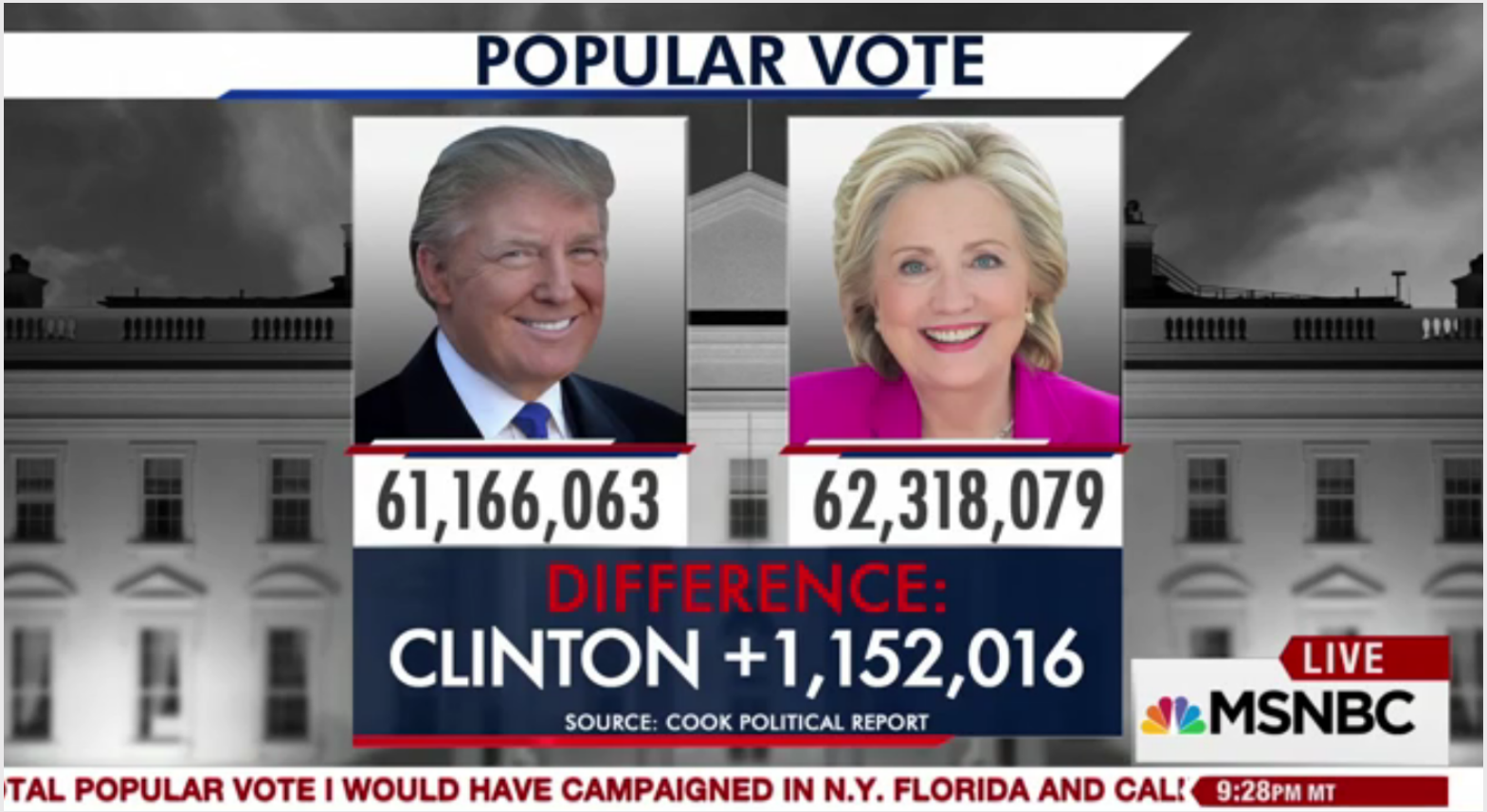


- Dictators have to balance power-sharing with control over the population
- Broadly:
  - Monarchies and military dictatorships tend to emphasize control
  - Civilian dictatorships tend to emphasize power-sharing

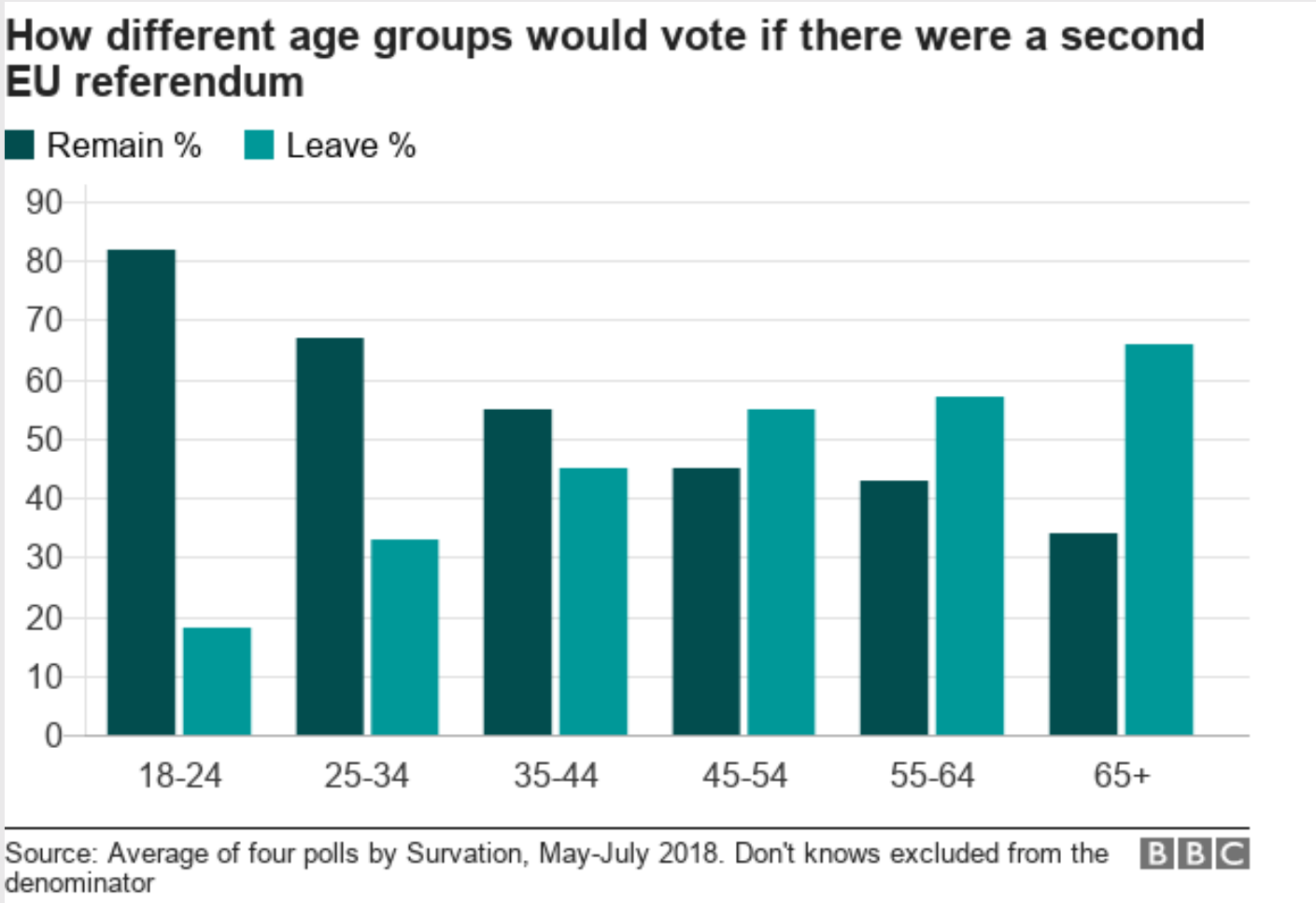
Democracy is about majority rule

- Some people like **democracy** because it is a **fair** way to make decisions
- We often assume **fairness** $\rightarrow$ reflecting the **preferences of the majority**
- A fair choice is that preferred by most people

So this is unfair



How about this?



A big assumption

- Sometimes it seems obvious that what the majority prefers is fair
- Sometimes, the **majority may end up making decisions against the desires of a minority**
- This is a problem if the **minority is directly affected** but the majority is not
- We will **assume that pro-majority outcomes are fair** for class purposes, but keep this in mind!

Social choice theory

- The study of **collective decision** processes and procedures
- Not a single theory, but an array of useful models to understand decision-making and its challenges
- Assume that decision makers in a democracy are **rational actors**
- **Rational: Complete** and **transitive** preferences

Complete preferences

- An actor has **complete preferences** if they can compare any pair of options and determine if they prefer one, the other, or are indifferent
  - Imagine we have options x, y, and z
  - If I grab x and y. I can tell which one I like better (or whether I am indifferent)
  - If I can do the same for every pair of options, then my preferences are complete

Transitive preferences

- An actor has **transitive preferences** if their preferences are not cyclical
  - $\geq$ means **weak preference** `(at least as good as)`
  - If $x \geq y$ AND $y \geq z$
  - Then $x \geq z$
  - If this is true, individual preferences are transitive

Condorcet's Paradox

- A group composed from rational individuals does not necessarily have rational preferences as a collective
- This means that even **a group of competent decision makers** can reach **socially sub-optimal decisions** `(want it or not)`
- **Why?**

An example

- A city council made of **three individuals** has to decide whether to:

  1. Increase social services: I
  2. Decrease social services: D
  3. Maintain current levels of services: C

Note: Three individuals is the smallest group in which it is meaningful to have decision making rules.

Preferences

- Each council member has **complete** and **transitive** preferences

Left	Center	Right
$I > C > D$	$C > D > I$	$D > I > C$

- Assume they require majority rule to reach a decision
- Imagine they select along alternatives via **round-robin**
- **Round-robin:** A tournament in which each competitor plays in turn against every other
- In this case, councilors will vote on every possible combination of alternative pairs and we will see which alternative wins more rounds

Round-robin results

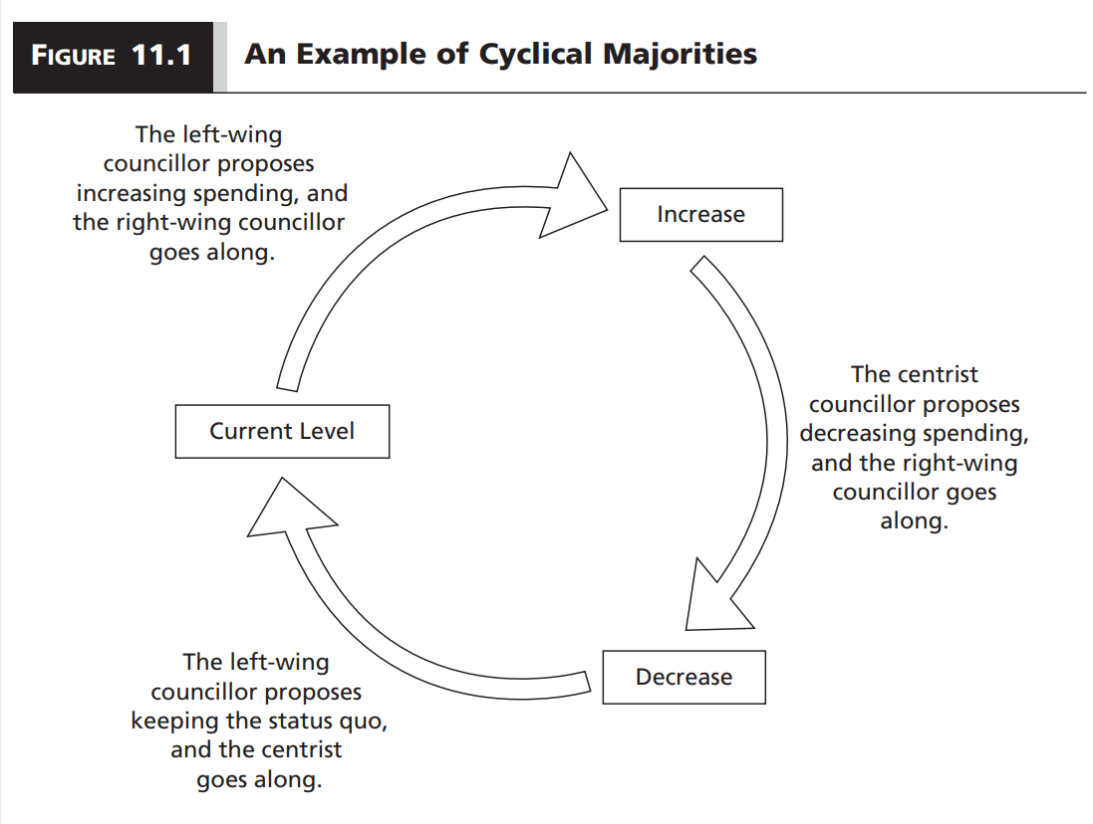
Left	Center	Right
$I > C > D$	$C > D > I$	$D > I > C$

Round	Contest	Winner	Majority
1	I vs D	D	Center and Right
2	C vs I	I	Left and Right
3	C vs D	C	Left and Center

- Group preference: $D > I > C > D$. **Intransitive!**
- There is no **Condorcet winner**
  - A candidate `(alternative)` that wins in a two-candidate election against any other candidate

Condorcet's Paradox in practice

- Round-robin elections are rare. But beyond the game, the **paradox can lead to cyclical majorities** over time



But cycles like this are an exception

- We do see democracies making somewhat coherent, perhaps rational, decisions
- Why is this the case? How do we overcome Condorcet's paradox?
- Two reasons:
    1. Preference orderings
    2. Decision making rules

Preference orderings

- The Right councilor is weird in that they prefer $D > I > C$
- Let's change their preference to a mirror of the Left councilor: $D > C > I$

Left	Center	Right
$I > C > D$	$C > D > I$	$D > C > I$

Round	Contest	Winner	Majority
1	I vs D	D	Center and Right
2	C vs I	C	Center and Right
3	C vs D	C	Left and Center

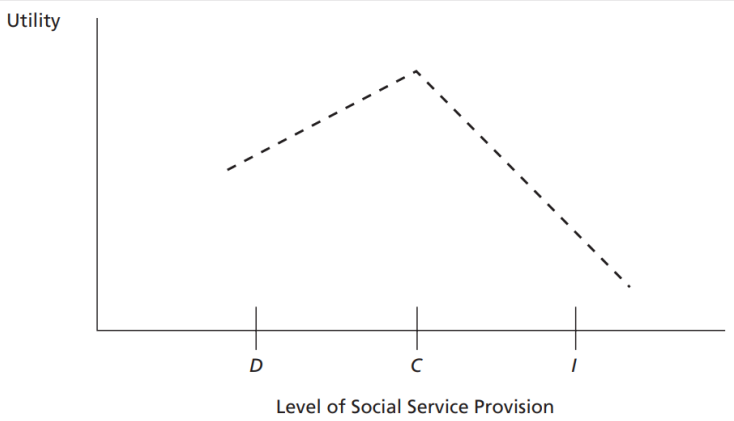
- Now C is a **Condorcet winner**

Preference orderings

- Condorcet's paradox only says that **it is possible** for individual rational preferences to aggregate into irrational preferences
- But some **"well-behaved"** preference orderings can find a Condorcet winner

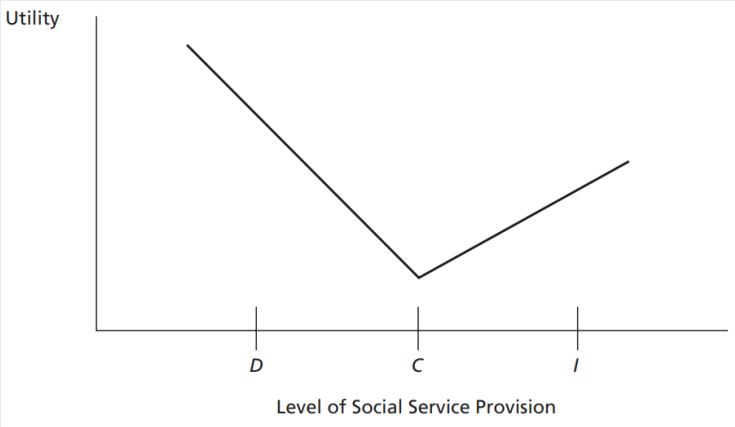
Single-peaked preferences

Center councilor



✔ Single-peaked

Right councilor



✖ Not single-peaked

- We say that the right councilor has "weird" preferences because they are **not single-peaked**

Note: I may ask you to tell me whether preferences are single-peaked or not in the quiz.

Median voter theorem

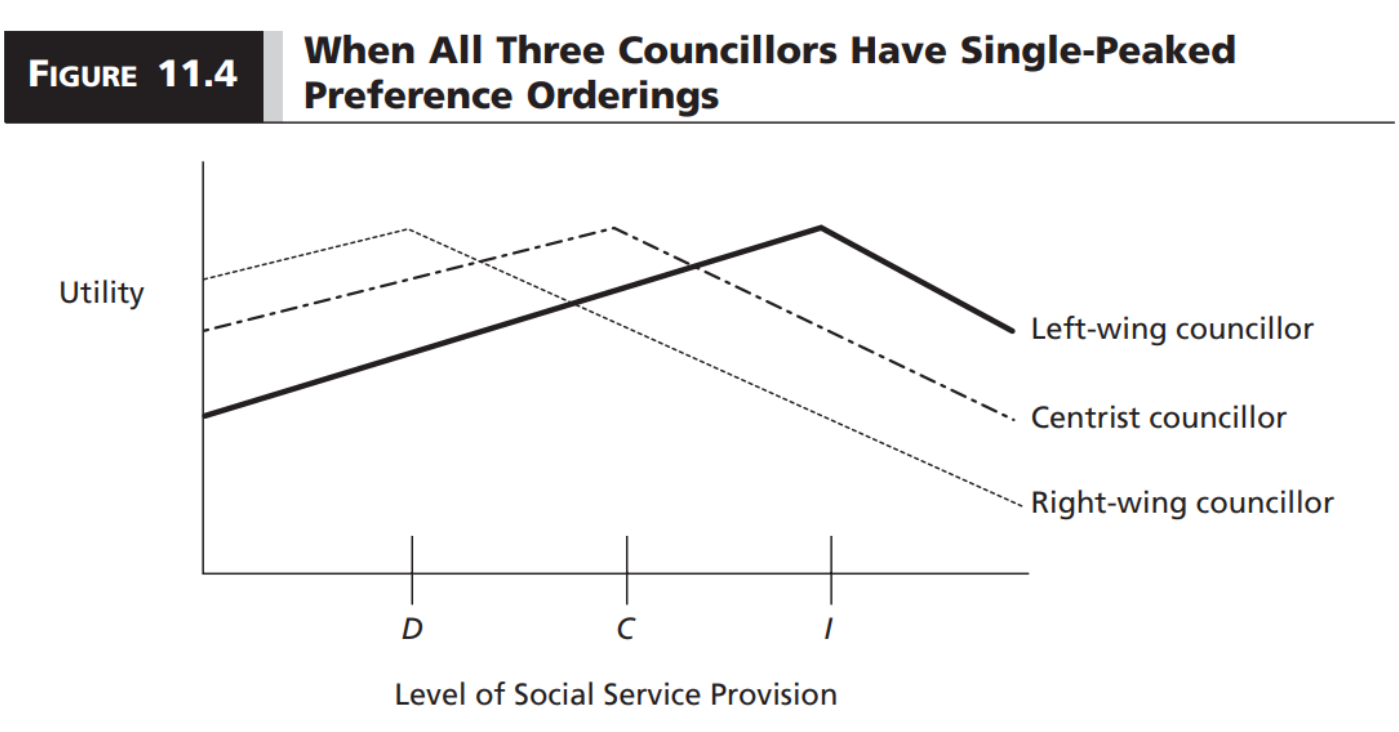
- If we can assume the following:

  1. The number of voters is odd
  2. Voter preferences are single-peaked
  3. We can summarize preferences in one dimension
  4. Voters vote sincerely

- Then the ideal point of the **median voter** will win against any alternative in a pair-wise majority rule election

- So the ideal point of the median is the **Condorcet winner**

Visualizing the median voter theorem



- Center can get their preferred option by joining either Left or Right against the other
- No Condorcet's paradox here

Problems with median voter theorem

- Nothing wrong with not-single peaked preferences
- Many political issues are inherently multi-dimensional
- **Chaos theorem:**
  - $\text{Dimensions} \geq 2 + \text{Voters} \geq 3 = \text{No Condorcet winner}$
  - Unless we have a rare distribution of ideal points

Note: Make sure you understand the two-dimensional voting figures in the book

How often are we in trouble?

TABLE 11.3 **Proportion of Possible Strict Preference Orderings without a Condorcet Winner**

Number of alternatives	Number of voters: 3	5	7	9	11 →	Limit
3	0.056	0.069	0.075	0.078	0.080	0.088
4	0.111	0.139	0.150	0.156	0.160	0.176
5	0.160	0.200	0.215			0.251
6	0.202					0.315
↓						↓
Limit	1.000	1.000	1.000	1.000	1.000	1.000

Assuming no "weird" preferences. The proportion of strict preference orderings without a Condorcet winner increases with the number of voters and alternatives.

In practice

- Many real political decisions involve bargaining, which implies an infinite number of alternatives
- Also, most meaningful decision making bodies have more members than a handful
- In general, we cannot rely on majority rule to produce a coherent sense of what the group wants
- We need some way to narrow down alternatives
- **We need decision-making rules**

Decision-making rules

- Rules can establish **how to decide** or **who is in charge of leading decision-making**

  1. The Borda count
  2. An agenda setter

Borda count

- Rational individuals have complete, transitive preferences
- Then why do we ask them to cast a vote for one alternative?
- We could instead ask them to rank all alternatives
- The **Borda count** does this!
- Each individual assigns points to alternatives according to their preference ranking
- The alternative with most points wins

Using the Borda count

Alternative	Left	Center	Right	Total
Increase spending	3	1	2	**6**
Decrease spending	1	2	3	**6**
Current spending	2	3	1	**6**

- The Borda count did not help 😣
- Wanna make it worse? Let's introduce an **irrelevant alternative**

An irrelevant alternative

- Let's say another voting option is future cuts in social service provision: FC
- Preferences now look like this:

Left	Center	Right
$I > C > D > FC$	$C > D > FC > I$	$D > FC > I > C$

- You will see why FC is irrelevant in a second...

Borda count with irrelevant alternatives

Alternative	Left	Center	Right	Total
Increase spending	3	0	1	**4**
Decrease spending	1	2	3	**6**
Current spending	2	3	0	**5**
Future cuts	0	1	2	**3**

- FC is irrelevant in that it does not stand a chance against any of the other alternatives in round-robin `(I am not showing you this but I may ask you about it)`
- Introducing a **seemingly** irrelevant alternative now gives us a clear winner
- Politicians can manipulate decisions rules that are not **independent of irrelevant alternatives**
- Rather than argue for what you prefer the most, you can introduce an irrelevant alternative

Agenda setting

- If decision rules do not work, perhaps we need someone to control the agenda
  - **First round:** I vs D
  - **Second round:** Winner of first round vs. C
- Some real-world legislatures operate like this

Pairwise vote under different agendas

Left	Center	Right
$I > C > D$	$C > D > I$	$D > I > C$

Agenda	Round 1 vote	Round 1 winner	Round 2 vote	Round 2 winner
1	I vs D	D	D vs C	C
2	C vs I	I	I vs D	D
3	C vs D	C	C vs I	I

- Whatever alternative is introduced in the second round wins
- The agenda setter can get their preferred outcome
- **A powerful agenda setter is a dictator!**

Summary

- Unless we are lucky to have individual actors whose preferences do not lead to cyclical majorities, then either of two things will happen:

  1. Decision-making leads to unstable policy outcomes

  2. An agenda setter can exploit the decision-making rules to get their most preferred outcome

- Does that mean we should drop majority rule in a democracy?

Arrow's impossibility theorem

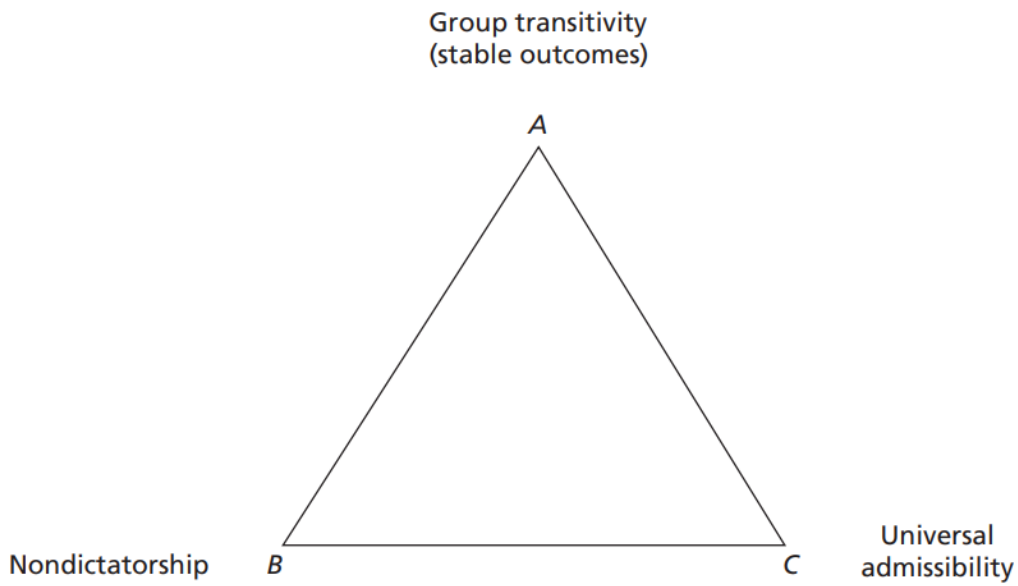
- Four **fairness conditions**:

  1. **Non-dictatorship:** No individual can fully determine the outcome
  2. **Universal admissibility:** Individuals can adopt any rational preference ordering
  3. **Unanimity (pareto optimality):** If everyone prefers $x > y$, the then group should not choose y if x is available
  4. **Idependence of irrelevant alternatives (IIA):** Introducing irrelevant alternatives does not affect the outcome

- We normally assume **3** and **4** are uncontroversial. We want better outcomes for society and we want to weed out irrelevant alternatives
- The hard part is to balance the first two

You have to choose a side

FIGURE 11.10 **Arrow's Institutional Trilemma**



- Ensuring one of these conditions requires sacrificing the other two

The impossibility in practice

- Any **seemingly smooth** decision making process has at least one of the following:
  - Unstable majorities
  - Someone takes over the process
  - Some alternatives (and thus people who prefer them) are excluded
- **It is virtually impossible to have a decision rule without leaving someone unhappy**

Takeaways

- We believe democracy is fair because of majority rule
- But nearly all decision-making rules using majority have issues
- Things only work well in the rare case of well-behaved preferences
- **To function properly, a democracy has to systematically exclude a non-trivial portion of the population**

Reminder:

News Report 4 Due Friday 5:00 PM

Next Week:

Democratic Systems